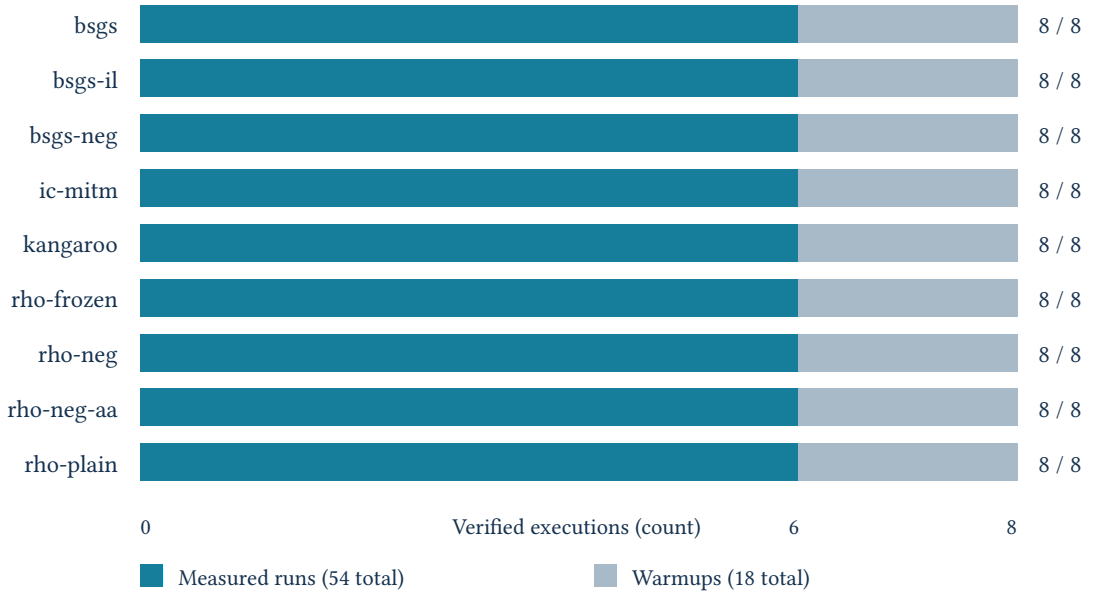
Real crypto on EKS: correctness and S3 persistence

2026-10-19 · Session ECBS1hc931106fd275

Result: 72/72 elliptic-curve solver executions verified. All 54 measured runs reproduced in both the pod audit and a second audit of the S3 download on the harness host. Neither audit reported a problem.

Verified crypto computations by solver arm

Session ECBS1hc931106fd275 · 2 targets · 3 measured rounds + 1 warmup



Source: counts.json, derived by counting verified records in artifacts/session/records.jsonl.

Workload. Existing public synthetic 20-bit prime-field fixture; modulus 563153, subgroup order 562427. Two targets, three measured rounds and one warmup across nine arms. Every arm recovered the known scalars 330146 and 138031. Verification checked each equation [k]G = Q.

Provenance. Codex MCP Job ecbench-real-20261019-1922 in EKS isolab, us-east-2. Spot c8in.xlarge, Intel Xeon 6975P-C; 2 requested vCPUs, 4 GiB. Native ecbench source cb2ec6873f30b683f7885b53c410d69965f02c20. Full image digest, inputs and exact commands: ../../protocol.md and ../../request.json.

Persistence. All 17 S3 objects were downloaded. Every one of the 16 payload files matched the SHA-256 manifest produced inside the pod. The sealed session was then fully replayed outside the worker pod using the same frozen native binary.

```
s3://isolab-results-104186811093-us-east-2/
runs/ecbench-real-20261019-1922/
5319a3b0-ba85-44cc-be19-02d123364899/
```

Evidence. ../../artifacts/session/records.jsonl; ../../artifacts/audit.json; ../../host-replay.json; ../../cloud-receipt.json. Graph data: ../../counts.json (verified records grouped by arm). Counts are complete observations; no uncertainty interval applies.

Second audit SHA-256: `386d52abd0a8a36f32f787051256d5c88e5e7629bbfba595e07085b14a90ea82`

Limits. All records earned L0: the quiet-start and host-eviction requirements for stronger isolation were unmet. No timing comparison, speedup, deployed-curve security claim or strict-fidelity result is reported. The second audit is a separate execution of the same implementation. This check adds no new curve or performance finding; canonical scoreboards and curve diagrams are unchanged.